Naive Bayes Classifier:.

Understanding Bayesian Classifiers and Bayes Theorem

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Why Probabilistic Classification?

In many real-world problems, the relationship between **attributes (features)** and the **class label** is **not perfectly predictable** — it's *non-deterministic*.

That means:

Even if two records have identical attribute values, they might still belong to different classes.

Examples:

Heart Disease Prediction

- Attributes: Diet, Exercise frequency
- Class: Risk of heart disease (Yes/No)

Even if two people eat healthily and exercise:

- One might still develop heart disease (due to heredity or smoking).
- Another might not.

This happens because of **hidden factors** or **noise** not captured in the dataset.

In Short

- Real-world data often involves **uncertainty**.
- To model such uncertainty, we use **probabilistic approaches** instead of deterministic ones.

Understanding Bayesian Classifiers and Bayes Theorem

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- Naïve Bayes algorithm is a supervised learning algorithm, which is based on Bayes theorem and used for solving classification problems.
- It is mainly used in text classification that includes a high dimensional training dataset.
- Naïve Bayes Clasifier is one of the simple and most effective Classification algorithms which helps in building the fast machine learning models that cam make quick predictions.
- It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.
- Some popular examples of Naïve Bayes Algorithm are spam filtration , Sentimental analysis and classifying articles

Understanding Bayesian Classifiers and Bayes Theorem

- It is called Naïve because it assumes that the occurrence of a certain feature is independent of the occurrence of other features.
- Such as if the fruit is identified on the bases of color, shape and taste then red, spherical and sweet fruit is recognized as an apple.
- Hence each feature individually contributes to identify that it is an apple without depending on each other.
- It is called Bayes because it depends on the principle of Bayes Theorem

Bayes Theorem

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Bayes Theorem gives the conditional probability of an event X given another event Y has occurred. Where

$$P(Y|X) = \frac{P(X|Y) \times P(Y)}{P(X)}$$

P(Y|X) = Conditional Probability of Y given X
P(X|Y) = Conditional Probability of X given Y
P(X) = Probability of event X
P(Y) = Probability of event Y

Algorithm steps

Step 1 : Compute the Prior Probabilities for each of the class of fruits

Step 2 : Compute the probability of evidence that goes in the denominator

Step 3 : Compute the probability of likelihood of evidences that goes in the numerator.

Step 4 : Substitute all the 3 equations into Naïve Bayes formula , to get the probability that it is a banana.

Joint Probability

- The joint probability is the probability of two (or more) simultaneous events, often described in terms of events A and B from two dependent random variables, e.g. X and Y.
- The joint probability is often summarized as just the outcomes, e.g. A and B.

– Joint Probability: Probability of two (or more) simultaneous events, e.g. P(A and B) or P(A,B).

Conditional Probability

- The conditional probability is the probability of one event given the occurrence of another event, often described in terms of events A and B from two dependent random variables e.g. X and Y.

– Conditional Probability: Probability of one (or more) event given the occurrence of another event, e.g. P(A given B) or P(A | B).

Conditional Probability

- The joint probability can be calculated using the conditional probability; for example: – P(A, B) = P(A | B) * P(B)
- This is called the product rule. Importantly, the joint probability is symmetrical, meaning that: – P(A, B) = P(B, A)
- The conditional probability can be calculated using the joint probability; for example: – P(A | B) = P(A, B) / P(B)
- The conditional probability is not symmetrical; for example:
  – P(A | B) != P(B | A)

Conditional Probability

- Specifically, one conditional probability can be calculated using the other conditional probability; for example:
  - P(A|B) = P(B|A) * P(A) / P(B)
- The reverse is also true; for example:
  - P(B|A) = P(A|B) * P(B) / P(A)
- This alternate approach of calculating the conditional probability is useful either when the joint probability is challenging to calculate (which is most of the time), or when the reverse conditional probability is available or easy to calculate.

Bayes Theorem

- Bayes Theorem: Principled way of calculating a conditional probability without the joint probability. It is often the case that we do not have access to the denominator directly, e.g. P(B).
- We can calculate it an alternative way; for example:
  - P(B) = P(B|A) * P(A) + P(B|not A) * P(not A)
- This gives a formulation of Bayes Theorem that we can use that uses the alternate calculation of P(B), described below:
  - P(A|B) = P(B|A) * P(A) / P(B|A) * P(A) + P(B|not A) * P(not A)

Bayes Theorem

- Firstly, in general, the result P(A|B) is referred to as the posterior probability and P(A) is referred to as the prior probability.
  - P(A|B): Posterior probability.
  - P(A): Prior probability.
- Sometimes P(B|A) is referred to as the likelihood and P(B) is referred to as the evidence.
  - P(B|A): Likelihood.
  - P(B): Evidence.
- This allows Bayes Theorem to be restated as:
  - Posterior = Likelihood * Prior / Evidence

Naïve Bayes Classifier

- Naive Bayes classifiers are a collection of classification algorithms based on Bayes' Theorem.
- It is not a single algorithm but a family of algorithms where all of them share a common principle, i.e. every pair of features being classified is independent of each other.

Understanding Bayesian Classifiers and Bayes Theorem

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Bayes Theorem: Foundation of Probabilistic Learning

Bayes theorem helps us **combine prior knowledge** with **new evidence** from data to make better predictions.

Let's Understand Through an Example

Scenario: Two football teams – **Team 0** and **Team 1** – play against each other.

Event	Probability
Team 0 wins overall	0.65
Team 1 wins overall	0.35
Team 0 wins while Team 1 hosts	0.30
Team 1 wins while hosting	0.75

Question:

If **Team 1** is hosting the next match, which team is more likely to win?

1. What is Y? (The Event We Want to Predict)

In Bayes Theorem,

Y = the class label or outcome we want to predict.

In this football example:

- **Y = 1 → Team 1 wins**
- **Y = 0 → Team 0 wins**

So Bayes is helping us answer:

"What is the probability that Team 1 wins, given that Team 1 is hosting?"

i.e., $P(Y=1 \mid X=1)$

2. What is X? (The Evidence / Observed Condition)

X is the observed evidence or feature.

Here:

- **X = 1 → Team 1 is hosting the match**
- **X = 0 → Team 1 is NOT hosting (Team 0 is hosting)**

So **X is not the winner** — it is the **condition** under which the match happens.

We are conditioning on:

"Team 1 is hosting (X = 1)"

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Using Bayes Theorem

Bayes Theorem formula:

$$P(Y|X) = \frac{P(X|Y) \times P(Y)}{P(X)}$$

- Y = class variable
- X = vector of features
- $P(Y)$ = prior probability of class Y
- $P(X \mid Y)$ = likelihood of features given class
- $P(X)$ = evidence (normalizing factor)

Where:

- **Y:** Event we want to predict (Winner)
- **X:** Observed evidence (Team hosting)

We need **P(Y=1|X=1)** → probability that *Team 1 wins given that Team 1 hosts.*

$$P(Y=1|X=1) = \frac{P(X=1|Y=1)P(Y=1)}{P(X=1|Y=1)P(Y=1) + P(X=1|Y=0)P(Y=0)}$$

$$= \frac{0.75 \times 0.35}{(0.75 \times 0.35) + (0.30 \times 0.65)} = 0.5738$$

- $P(Y=1)$ = Prior probability of class 1
- $P(Y=0)$ = Prior probability of class 0
- $P(X=1|Y=1)$ = Likelihood of X=1 given class = 1
- $P(X=1|Y=0)$ = Likelihood of X=1 given class = 0

So,

P(Team 1 wins | Team 1 hosts) = **0.5738**

P(Team 0 wins | Team 1 hosts) = **0.4262**

The Denominator Part

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$$P(X) = P(X \mid Y)\,P(Y) \;+\; P(X \mid \neg Y)\,P(\neg Y)$$

This is called the **Law of Total Probability**.

Think in terms of ALL ways X can happen

Suppose you have an event X (like:

- *Team 1 is hosting*, or
- *Test result is positive*)

X can happen in TWO possible worlds:

World 1: Y happens

(e.g., disease present or Team 1 wins)

World 2: Y does NOT happen

(no disease or Team 1 loses)

These two cover *ALL* possibilities.

So the probability X happens must be:

Probability X happens AND Y happened

plus

Probability X happens AND Y did not happen

Use the multiplication rule (The **multiplication rule** is a fundamental probability rule that tells you how to find the probability of **two events happening together** (their intersection).)
The probability X happens **and** Y happens:

$$P(X \cap Y) = P(X|Y)P(Y)$$

Similarly:

$$P(X \cap \neg Y) = P(X|\neg Y)P(\neg Y)$$

Add the two possibilities

Because either:

- X happens with Y
- OR X happens with not-Y

These cases are *mutually exclusive*.

So we add them:

$$P(X) = P(X \cap Y) + P(X \cap \neg Y)$$

Now substitute the multiplication rule:

$$P(X) = P(X|Y)P(Y) + P(X|\neg Y)P(\neg Y)$$

You now have:

$$P(X) = P(X|Y)P(Y) + P(X|\neg Y)P(\neg Y)$$

This is exactly the denominator in Bayes.

- A probabilistic framework for solving classification problems
- Conditional Probability:

$$P(Y \mid X) = \frac{P(X,Y)}{P(X)}$$

$$P(X \mid Y) = \frac{P(X,Y)}{P(Y)}$$

- Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

Bayes Classifier-Example

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Suppose you have invited two of your friends Alex and Martha to a dinner party. You know that Alex attends 40% of the parties he is invited to. Further, if Alex is going to a party, there is an 80% chance of Martha coming along. On the other hand, if Alex is not going to the party, the chance of Martha coming to the party is reduced to 30%. If Martha has responded that she will be coming to your party, what is the probability that Alex will also be coming?

1) Define the events

- A= 1: Alex goes to the party
- A = 0: Alex does not go
- M = 1: Martha goes to the party

2) Known probabilities

Prior probability of Alex going: $P(A = 1) = 0.4$

$P(A = 0) = 0.6$)since total probability = 1)

Conditional probabilities of Martha going:

If Alex goes: $P(M = 1 \mid A = 1) = 0.8$

If Alex doesn't go: $P(M = 1 \mid A = 0) = 0.3$

Bayes Classifier-Example

3) Probability that Alex goes to the party provided Martha is going is: (applying Bayes Theorem)

$$P(A = 1 \mid M = 1) = \frac{P(M = 1 \mid A = 1) \cdot P(A = 1)}{P(M = 1)}$$

To compute P(M=1) using law of total probability, we have:

$$P(M = 1) = P(M = 1 \mid A = 1) \cdot P(A = 1) + P(M = 1 \mid A = 0) \cdot P(A = 0)$$

Substituting values:

$P(M=1) = (0.8 * 0.4) + (0.3 * 0.6) = 0.50$

4) Substituting this into Bayes' formula:

$P(A=1|M=1) = (0.8 * 0.4)/ 0.5 = 0.64$

If Martha is going to the party, there's a 64% chance Alex will go too—even though his original chance was only 40%.

Problem 2

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A disease affects 1% of a population. A diagnostic test has 95% sensitivity (true positive rate) and 90% specificity (true negative rate). A patient tests positive. What is the probability they actually have the disease?

Problem 3

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From historic data, 40% of emails are spam. If an email contains the word “discount”, that word appears in 60% of spam and 10% of non-spam. You observe “discount”. What’s the probability the email is spam?

Problem 4

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5% of applicants default historically. A credit bureau score gives a “risky” flag. Among defaulters, 70% get the risky flag; among non-defaulters 20% get the risky flag. Applicant flagged risky — what's default probability?

Problem 5

A machine produces 2% defective units. A vibration sensor reading is high for 85% of defective units and for 5% of non-defective units. You observe high vibration. What's chance the unit is defective?

Problem 6

Problem: Two sensors A and B monitor equipment. Fault occurs with prior $P(F) = 0.04$. Sensor A alarms with $P(A = 1 \mid F) = 0.9$, $P(A = 1 \mid \neg F) = 0.05$. Sensor B alarms with $P(B = 1 \mid F) = 0.85$, $P(B = 1 \mid \neg F) = 0.10$. Assume conditional independence of sensors given fault. Both sensors alarm. What's $P(F \mid A = 1, B = 1)$?

Bayes Classifier-Example Data

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Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$$

- We need to estimate
- P(Evade = Yes | X) and P(Evade = No | X)
  - In the following we will replace

    Evade = Yes by Yes, and Evade = No

    by No

Bayes Classifier-Example Data

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10	No	Single	90K	Yes

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$$

Using Bayes Theorem:

- $P(\text{Yes} \mid X) = \dfrac{P(X \mid \text{Yes})P(\text{Yes})}{P(X)}$
- $P(\text{No} \mid X) = \dfrac{P(X \mid \text{No})P(\text{No})}{P(X)}$
- How to estimate $P(X \mid \text{Yes})$ and $P(X \mid \text{No})$?

Bayes Classifier-Example Data

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Tid	Refund	Marital Status	Taxable Income	Evade
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8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given a Test Record:

$$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$$

$$P(X \mid \text{Yes}) = P(\text{Refund} = \text{No} \mid \text{Yes}) \times P(\text{Divorced} \mid \text{Yes}) \times P(\text{Income} = 120\text{K} \mid \text{Yes})$$

$$P(X \mid \text{No}) = P(\text{Refund} = \text{No} \mid \text{No}) \times P(\text{Divorced} \mid \text{No}) \times P(\text{Income} = 120\text{K} \mid \text{No})$$

Estimate Probabilities from Data

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Tid	Refund	Marital Status	Taxable Income	Evade
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4	Yes	Married	120K	No
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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- P(y) = fraction of instances of class y
  - e.g., P(No) = 7/10,
    P(Yes) = 3/10
- For categorical attributes:

$$P(X_i = c|y) = \frac{n_c}{n}$$

  $P(X_i = c| y) = n_c / n$
  - where $|X_i = c|$ is number of instances having attribute value $X_i = c$ and belonging to class y
  - Examples:

    P(Status=Married|No) = 4/7

    P(Refund=Yes|Yes)=0